

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security

Practical 1 B

1B AIM : Design and implement algorithms to encrypt and decrypt messages using transposition techniques cli

Code :

```
def rail(t, k, d=False):
```

```
    if not d:
        r = [""] * k
        i, step = 0, 1
        for ch in t:
            r[i] += ch
            i += step
            if i in (0, k - 1):
                step *= -1
        return "".join(r)
```

```
    p = [[] for _ in range(k)]
    i, step = 0, 1
    for j in range(len(t)):
        p[i].append(j)
        i += step
        if i in (0, k - 1):
            step *= -1
```

```
    res = [""] * len(t)
    idx = 0
    for row in p:
        for pos in row:
            res[pos] = t[idx]
            idx += 1
    return "".join(res)
```

```
def col(t, k, d=False):
```

```
    n = len(k)
    order = sorted(range(n), key=lambda i: k[i])
```

```
    if not d: # Encrypt
        cols = [""] * n
```

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```
for i, ch in enumerate(t):
    cols[i % n] += ch
return "".join(cols[i] for i in order)
```

```
l = len(t)
size = [l // n + (i < l % n) for i in range(n)]
cols, idx = {}, 0
```

```
for i in order:
    cols[i] = t[idx:idx + size[i]]
    idx += size[i]
```

```
return "".join(
    cols[j][i]
    for i in range(max(size))
    for j in range(n)
    if i < len(cols[j])
)
```

```
msg = input("Msg: ")
```

```
print("1. Rail Fence\n2. Columnar")
ch = input("Choose: ")
```

```
if ch == "1":
    k = int(input("Rails: "))
    e = rail(msg, k)
    print("Enc:", e)
    print("Dec:", rail(e, k, True))
```

```
elif ch == "2":
    key = input("Key: ")
    e = col(msg, key)
    print("Enc:", e)
    print("Dec:", col(e, key, True))
```

```
else:
    print("Invalid choice")
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```

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Output

```
C:\Users\IT-50>"C:\Users\IT-50\AppData\Local\Programs\Python\Python314\cybersecurity prcatial 2a.py"
Msg: hello
1. Rail Fence
2. Columnar
Choose: 2
Key: carss
Enc: ehlllo
Dec: hello
C:\Users\IT-50>
```

```
C:\Users\IT-50>"C:\Users\IT-50\AppData\Local\Programs\Python\Python314\cybersecurity prcatial 2a.py"
Msg: adarsh shukla
1. Rail Fence
2. Columnar
Choose: 1
Rails: 5
Enc: ahdsua krhl sa
Dec: adarsh shukla
C:\Users\IT-50>
```

```
= RESTART: C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cybersecurity prcatial 2a.py
Msg: adarsh
1. Rail Fence
2. Columnar
Choose: 2
Key: adrssss
Enc: adarsh
Dec: adarsh
>>>
===== RESTART: C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cybersecurity prcatial 2a.py =====
Msg: adarsh
1. Rail Fence
2. Columnar
Choose: 2
Key: zebra
Enc: sadrah
Dec: adarsh
>>>
===== RESTART: C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cber security 2gui.py =====
>>>
===== RESTART: C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cybersecurity prcatial 2a.py =====
Msg: hello
1. Rail Fence
2. Columnar
Choose: 1
Rails: 3
Enc: hoel l
Dec: hello
>>>
```

Ln: 148 Col: 0

2 cm of rain Sat 09:54 13-07-2026

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1B AIM : Design and implement algorithms to encrypt and decrypt messages using transposition techniques gui

Code :

```
import tkinter as tk
from tkinter import messagebox
```

```
def rail(t, k, d=False):
    if k < 2: return t
```

```
    if not d:
        r = [""] * k
        i, step = 0, 1
        for ch in t:
            r[i] += ch
            i += step
            if i in (0, k - 1):
                step *= -1
        return "".join(r)
```

```
    p = [[] for _ in range(k)]
    i, step = 0, 1
    for j in range(len(t)):
        p[i].append(j)
        i += step
        if i in (0, k - 1):
            step *= -1
```

```
    res = [""] * len(t)
    idx = 0
    for row in p:
        for pos in row:
            res[pos] = t[idx]
            idx += 1
```

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```
return ".join(res)
```

```
def col(t, k, d=False):
    if len(k) < 2: return t

    n = len(k)
    order = sorted(range(n), key=lambda i: k[i])

    if not d: # Encrypt
        cols = [""] * n
        for i, ch in enumerate(t):
            cols[i % n] += ch
        return ".join(cols[i] for i in order)

    l = len(t)
    size = [l // n + (i < l % n) for i in range(n)]
    cols, idx = {}, 0

    for i in order:
        cols[i] = t[idx:idx + size[i]]
        idx += size[i]

    return ".join(
        cols[j][i]
        for i in range(max(size))
        for j in range(n)
        if i < len(cols[j])
    )
```

```
def process(mode):
    msg = entry_msg.get()
    key = entry_key.get()
    method = choice.get()

    if not msg:
        messagebox.showerror("Error", "Enter message")
    return
```

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```
try:
    if method == "Rail Fence":
        k = int(key)
        if k < 2:
            messagebox.showerror("Error", "Rails ≥ 2")
            return
        out = rail(msg, k, mode == "Dec")

    else:
        if len(key) < 2:
            messagebox.showerror("Error", "Key length ≥ 2")
            return
        out = col(msg, key, mode == "Dec")

    result.set(out)

except:
    messagebox.showerror("Error", "Invalid input")

root = tk.Tk()
root.title("Cipher Tool")
root.geometry("350x220")

tk.Label(root, text="Message").pack()
entry_msg = tk.Entry(root, width=35)
entry_msg.pack()

tk.Label(root, text="Key / Rails").pack()
entry_key = tk.Entry(root, width=35)
entry_key.pack()

choice = tk.StringVar(value="Rail Fence")
tk.OptionMenu(root, choice, "Rail Fence", "Columnar").pack(pady=5)

tk.Button(root, text="Encrypt", width=15, command=lambda: process("Enc")).pack(pady=2)
tk.Button(root, text="Decrypt", width=15, command=lambda: process("Dec")).pack(pady=2)
```

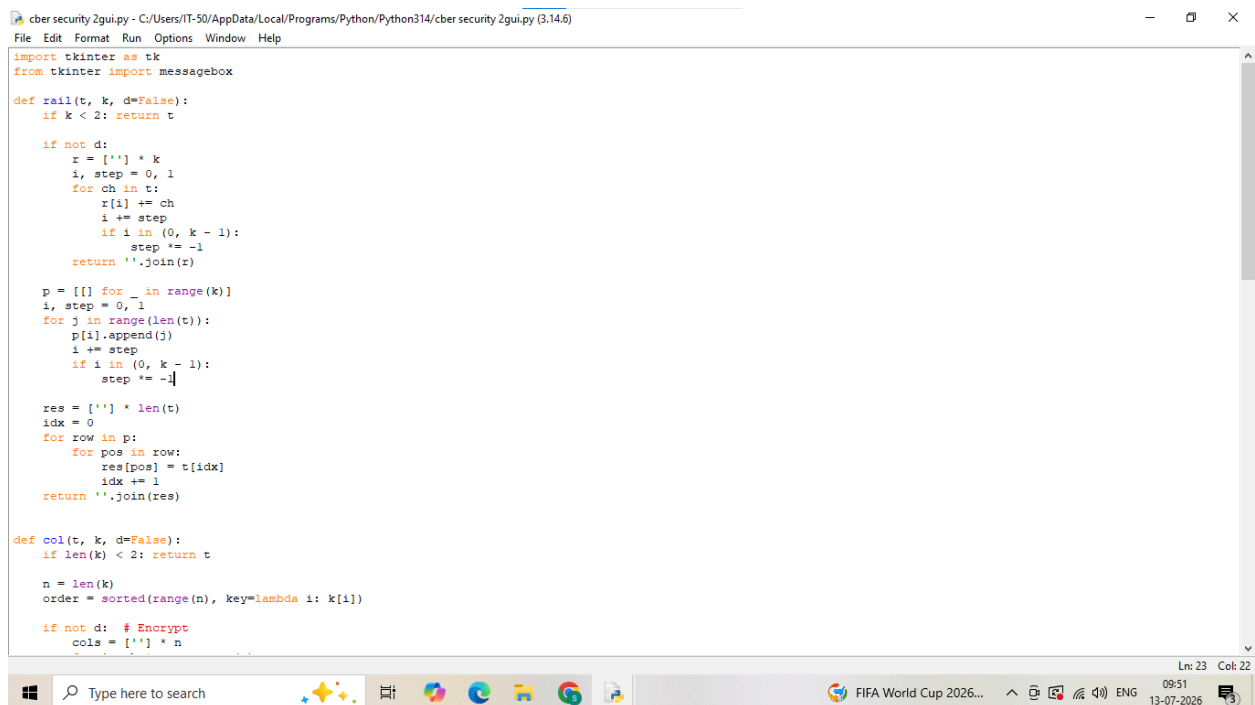
result = tk.StringVar()
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```
tk.Label(root, text="Result").pack()  
tk.Entry(root, textvariable=result, width=35).pack()  
  
root.mainloop()
```

Screen shot :



The screenshot shows a Python script editor window titled 'cber security 2gui.py'. The code defines two functions, 'rail' and 'col', which perform a rail fence cipher encryption. The 'rail' function takes a string 't' and a key 'k', and returns the encrypted string. The 'col' function takes a string 't' and a key 'k', and returns the encrypted string. The code is as follows:

```
import tkinter as tk  
from tkinter import messagebox  
  
def rail(t, k, d=False):  
    if k < 2: return t  
  
    if not d:  
        r = [''] * k  
        i, step = 0, 1  
        for ch in t:  
            r[i] += ch  
            i += step  
            if i in (0, k - 1):  
                step *= -1  
        return ''.join(r)  
  
    p = [[] for _ in range(k)]  
    i, step = 0, 1  
    for j in range(len(t)):  
        p[i].append(t[j])  
        i += step  
        if i in (0, k - 1):  
            step *= -1  
  
    res = [''] * len(t)  
    idx = 0  
    for row in p:  
        for pos in row:  
            res[pos] = t[idx]  
            idx += 1  
    return ''.join(res)  
  
def col(t, k, d=False):  
    if len(k) < 2: return t  
  
    n = len(k)  
    order = sorted(range(n), key=lambda i: k[i])  
  
    if not d: # Encrypt  
        cols = [''] * n  
        for i in range(len(t)):  
            for j in range(n):  
                cols[j] += t[i] + ' '
```

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```
cber security 2gui.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cber security 2gui.py (3.14.6)
File Edit Format Run Options Window Help

return ''.join(res)

def col(t, k, d=False):
    if len(k) < 2: return t

    n = len(k)
    order = sorted(range(n), key=lambda i: k[i])

    if not d: # Encrypt
        cols = [''] * n
        for i, ch in enumerate(t):
            cols[i % n] += ch
        return ''.join(cols[i] for i in order)

    l = len(t)
    size = [l // n + (i < l % n) for i in range(n)]
    cols, idx = [], 0

    for i in order:
        cols[i] = t[idx:idx + size[i]]
        idx += size[i]

    return ''.join(
        cols[j][i]
        for i in range(max(size))
        for j in range(n)
        if i < len(cols[j])
    )

def process(mode):
    msg = entry_msg.get()
    key = entry_key.get()
    method = choice.get()

    if not msg:
        messagebox.showerror("Error", "Enter message")
        return

    try:
        if method == "Rail Fence":
            k = int(key)
            if k < 2:
                messagebox.showerror("Error", "Rails ≥ 2")
                return
            out = rail(msg, k, mode == "Dec")
        else:
            if len(key) < 2:
                messagebox.showerror("Error", "Key length ≥ 2")
                return
            out = col(msg, key, mode == "Dec")
        result.set(out)
    except:
        messagebox.showerror("Error", "Invalid input")

root = tk.Tk()
root.title("Cipher Tool")
root.geometry("350x220")

tk.Label(root, text="Message").pack()
entry_msg = tk.Entry(root, width=35)
entry_msg.pack()

tk.Label(root, text="Key / Rails").pack()
entry_key = tk.Entry(root, width=35)

Ln: 23 Col: 22

cber security 2gui.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/cber security 2gui.py (3.14.6)
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def process(mode):
    msg = entry_msg.get()
    key = entry_key.get()
    method = choice.get()

    if not msg:
        messagebox.showerror("Error", "Enter message")
        return

    try:
        if method == "Rail Fence":
            k = int(key)
            if k < 2:
                messagebox.showerror("Error", "Rails ≥ 2")
                return
            out = rail(msg, k, mode == "Dec")
        else:
            if len(key) < 2:
                messagebox.showerror("Error", "Key length ≥ 2")
                return
            out = col(msg, key, mode == "Dec")
        result.set(out)
    except:
        messagebox.showerror("Error", "Invalid input")

root = tk.Tk()
root.title("Cipher Tool")
root.geometry("350x220")

tk.Label(root, text="Message").pack()
entry_msg = tk.Entry(root, width=35)
entry_msg.pack()

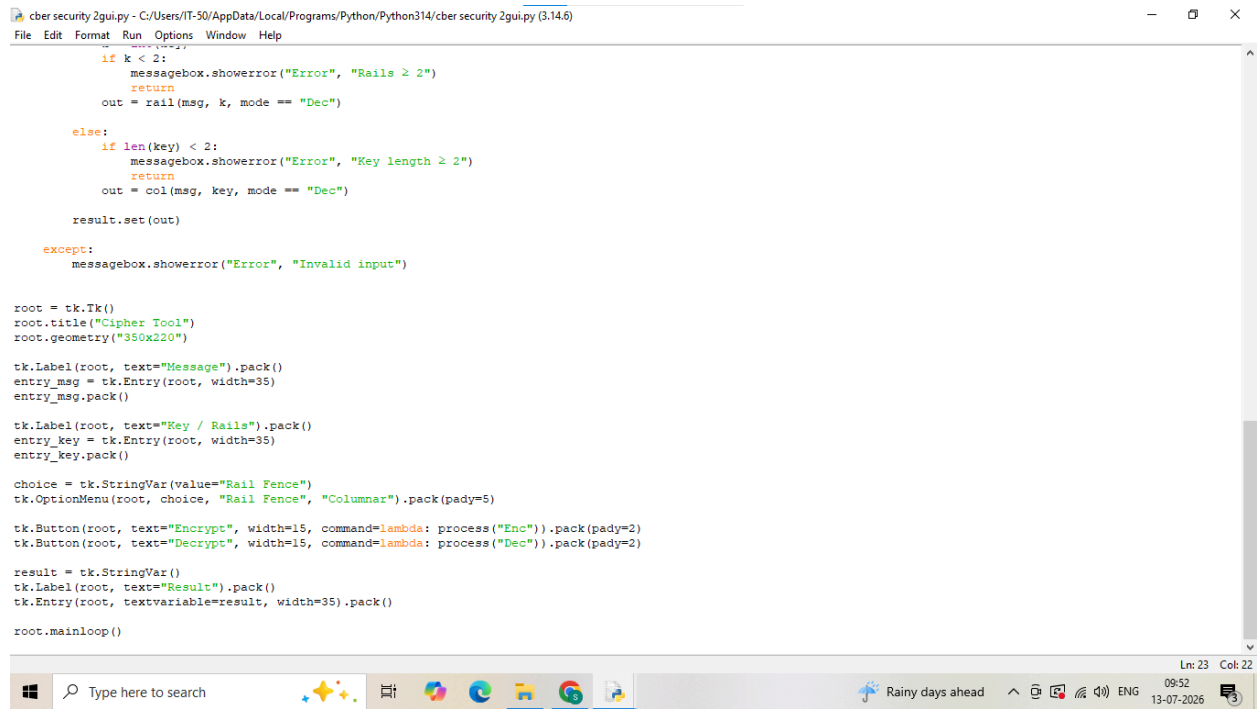
tk.Label(root, text="Key / Rails").pack()
entry_key = tk.Entry(root, width=35)

Ln: 23 Col: 22
```

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```
ciber security 2gui.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/ciber security 2gui.py (3.14.6)
File Edit Format Run Options Window Help

    if k < 2:
        messagebox.showerror("Error", "Rails ≥ 2")
        return
    out = rail(msg, k, mode == "Dec")

    else:
        if len(key) < 2:
            messagebox.showerror("Error", "Key length ≥ 2")
            return
        out = col(msg, key, mode == "Dec")

    result.set(out)

except:
    messagebox.showerror("Error", "Invalid input")

root = tk.Tk()
root.title("Cipher Tool")
root.geometry("350x220")

tk.Label(root, text="Message").pack()
entry_msg = tk.Entry(root, width=35)
entry_msg.pack()

tk.Label(root, text="Key / Rails").pack()
entry_key = tk.Entry(root, width=35)
entry_key.pack()

choice = tk.StringVar(value="Rail Fence")
tk.OptionMenu(root, choice, "Rail Fence", "Columnar").pack(pady=5)

tk.Button(root, text="Encrypt", width=15, command=lambda: process("Enc")).pack(pady=2)
tk.Button(root, text="Decrypt", width=15, command=lambda: process("Dec")).pack(pady=2)

result = tk.StringVar()
tk.Label(root, text="Result").pack()
tk.Entry(root, textvariable=result, width=35).pack()

root.mainloop()

Ln: 23 Col: 22
Type here to search
Rainy days ahead
09:52
13-07-2026
```

Output :

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The screenshot shows a web application interface for a Rail Fence cipher tool. It features a light gray background with several input fields and buttons. At the top, there is a label "Message" above a text input field containing the word "cars". Below this is a label "Key / Rails" above another text input field containing the word "ship". Underneath the key field is a dropdown menu currently set to "Columnar". Below the dropdown are two buttons: "Encrypt" and "Decrypt". At the bottom, there is a label "Result" above a text input field containing the word "arsc".

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The image shows a web application interface for a Rail Fence cipher. It consists of several input fields, buttons, and an output field, all arranged vertically on a light gray background.

- Message:** A text input field containing the word "cars".
- Key / Rails:** A text input field containing the word "ship".
- Columnar:** A checkbox that is currently checked.
- Encrypt:** A button to perform encryption.
- Decrypt:** A button to perform decryption.
- Result:** A text output field displaying the result "scar".

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